

ZEBULON ORACLE DATABASE — DUAL DATABASE MODE SETUP (Instructions for Replit Agent)

◆ PURPOSE

Zebulon must support both online and offline operation.

This requires two connected database layers:

1. Neon (Primary Online DB) – Used when internet is active
2. Local PostgreSQL (Offline Fallback) – Used when offline or disconnected

Both must mirror the same schema and be securely interchangeable.

✅ STEP 1 — IDENTIFY CURRENT DATABASE CONFIG

- Locate the existing DATABASE_URL pointing to the Neon instance.
- Confirm its use in:
 - .env
 - schema.prisma
 - All relevant services and server scripts

Do not modify the Neon connection. This remains the live database.

✅ STEP 2 — EXPORT BACKUP FROM NEON

- Export a full .sql backup of the Neon database.
- File must include:
 - All tables
 - Prisma metadata
 - Current content and settings
- Save as:
zebulon_oracle_backup.sql

✅ STEP 3 — INSTALL LOCAL POSTGRES

- Install a PostgreSQL server inside the Replit container.
- Create a local database with identical structure to Neon.

✅ STEP 4 — IMPORT BACKUP LOCALLY

- Load zebulon_oracle_backup.sql into the local Postgres database.
- Validate that all tables and Prisma fields match Neon.

✅ STEP 5 — CONFIGURE SWITCHING LOGIC

- Add a fallback detection in the Zebulon boot logic:
 - First check Neon DB availability.
 - If unreachable, auto-switch to the local DB connection.

- No user input should be required — this must be automatic.

✅ STEP 6 — ENVIRONMENT CONFIG

Set up the following:

DATABASE_URL_NEON=actual Neon connection string

DATABASE_URL_LOCAL=postgres://localhost/zebulon_oracle_local

System should prioritize DATABASE_URL_NEON, but fall back to DATABASE_URL_LOCAL on failure.

✅ STEP 7 — TEST BOTH MODES

Online Test:

- Confirm Neon is used
- Confirm live writes are visible in Neon dashboard

Offline Test:

- Disconnect internet
- Relaunch Zebulon
- Confirm modules operate normally using the local DB

✅ STEP 8 — OPTIONAL ADMIN EXPORT

Add logic in CONFIG panel (Admin only) to:

- Export latest Neon DB to .sql
- Import .sql into local fallback DB

This provides the Admin with manual sync capabilities.